How did you feel yesterday after our trip toVedanthangal? The answers would be:

(1) Very unhappy (2) Unhappy (3) Okay (4) Happy (5) Very happy

In each case, we know that number 5 is better than number 4 or number 3, but we don't know how much better it is. For example, is the difference between "Okay" and "Unhappy" the same as the difference between "Very Happy" and "Happy?" In fact we cannot say anything.

NOTE

In the 'ordinal scales' it is the order that matters, and that is all we get from these. It is to be noted that the mean cannot be computed from the ordinal data, but either median or mode can be computed.

Similarly, a medical practitioner can say the condition of a patient in the hospital as good, fair, serious and critical and assign numbers 1 for good, 2 for fair, 3 for serious and 4 for critical. The level of seriousness can be from 1 to 4 leading to $1 < 2$ or $2 < 3$ or $3 < 4$. However, the value here just indicates the level of seriousness of the patient only.

1.7.3 Interval scales:

In an interval scale one can also carryout numerical differences but not the multiplication and division. In other words, an interval variable has the numerical distances between any two numbers. For example, suppose we are given the following temperature readings in Fahrenheit: $60^\circ, 65^\circ, 88^\circ, 105^\circ, 115^\circ$, and 120°. It can be written that $105^\circ > 88^\circ$ or $60^\circ < 65^\circ$ which means that 105° is warmer than 88° and that 60° is colder than 65°. It can also be written that $65^\circ - 60^\circ = 120^\circ - 115^\circ$ because equal temperature differences are equal conveying the same amount of heat needed to increase the temperature from an object from 60° to 65° and from 115° to 125°. However it does not mean that an object with temperature 120° is twice as hot as an object with temperature 60°, though 120° divided by 60° is 2. The reason is converting the temperature in to Celsius we have:

$$60^\circ \text{F} = \frac{5}{9}[60^\circ - 32^\circ]C = 15.570^\circ C \text{ and}$$

$$120^\circ \text{F} = \frac{5}{9}[120^\circ - 32^\circ]C = 48.870^\circ C$$

NOTE

It is impossible to compute ratios without a real origin as zero.

In the above equations, it is clear from the left hand side that 120° F is twice of 60° F while the right hand side says 48.87° C is more than three times of $15.57^\circ C$. The reason for the difficulty is that the Fahrenheit and Celsius scales have artificial origins namely zeros(freezing point of centigrade measure is 0° C and the freezing point of Farenheit is 32° F) and there is no such thing as 'no temperature.'

Ch1_11th Stats.indd 12 2/3/2019 10:56:33 AM